Near-star spectral gaps on subdivided two-star qubit networks

Xiaojian Zeng
PBC School of Finance, Tsinghua University, 100190
xiaojianzeng@foxmail.com

Abstract

We determine the lowest excitation of an unweighted qubit network whose edges favor the unnormalized pair state $|00\rangle + s|11\rangle$. The graph consists of two hubs, each with k private leaves, joined through one shared vertex. For every integer $k \geq 100$ and every $\theta \in [1/2, 2]$, with $1 - s^2 = \theta/k$, the full ground and first excited energies are simple. Their gap satisfies $\Delta/(1-s^2) = 1 + 1/(2k) + (1-\theta)/(4k^2) + R$, with $|R| \leq 364/k^3$. In particular, the gap exceeds $1 - s^2$ throughout this domain and approaches that star-graph threshold uniformly on an infinite nonstar family. A full-star operator inequality controls all higher number sectors, while a separate Weyl bound excludes every internal competitor in the ground sector. Exact rational polynomial bounds and root isolation identify the two relevant eigenvalues and control the expansion. The limit has growing hub degree and fixed edge strength; it does not establish a bounded-degree thermodynamic result or resolve the sharp-gap conjecture for arbitrary graphs and parameters.

1 Model and relation to existing gap bounds

For a finite connected simple graph $G = (V, E)$ with one qubit per vertex, consider the time-independent closed-system Hamiltonian

$$H_s(G) = -\sum_{\{i,j\}\in E} (|00\rangle + s|11\rangle)(\langle 00| + s\langle 11|)_{ij}, \qquad 0 \leq s < 1. \tag{1}$$

Every term has identity spectators. The common edge energy is $J = 1$; for $J > 0$, all energies and gaps below scale by J. The parameter s is dimensionless. No edge phases, additional fields, weights, auxiliary systems, bath or drive are introduced. Connections and boundary conditions are specified entirely by G. The local ket is *unnormalized*: normalization would divide the spectrum and the gap by $1 + s^2$.

Wong, Bravyi, Gosset and Liu introduced this deformation and proved the star gap $1 - s^2$ for at least three vertices [1, Section 5.5, Lemma 13]. They conjectured $\Delta(H_s(G)) \geq 1 - s^2$ for all connected graphs and all $s < 1$ [1, Conjecture 3]. Here the full spectrum, with multiplicity, is ordered as $E_0 \leq E_1 \leq \cdots$, and $\Delta = E_1 - E_0$. The separate conjecture that the first excitation is odd is not assumed.

In the convention $Z|0\rangle = |0\rangle$, $Z|1\rangle = -|1\rangle$, the positive local term is

$$\frac{1+s^2}{4}(\mathbb{I} + Z_iZ_j) + \frac{1-s^2}{4}(Z_i + Z_j) + \frac{s}{2}(X_iX_j - Y_iY_j). \tag{2}$$

Thus the minimum-field theorem for Suzuki–Fisher Hamiltonians [2, Theorem 1] implies

$$\Delta(H_s(G)) \geq \frac{d_{\min}(G)}{2}(1 - s^2), \tag{3}$$

because the intrinsic field at vertex i is $d_i(1 - s^2)/4$. The sharp conjectured threshold already follows when $d_{\min} \geq 2$; the degree-one case retains a factor of two in this bound [3, Section 5.4].

Other Lee–Yang results provide an added-field gap bound [4, Theorem 11] and, for (1), the bound $(1 - s^2)/2$ [5, Theorem 6.10]. Positive-gap algorithmic results are therefore established prior work, rather than a consequence first obtained here. At the excluded endpoint $s = 1$, $H_1 = -2Q$, where Q is the positive sum of normalized EPR projectors studied by Ju and Nagda [6].

The present calculation determines the actual full gap, including its first relative corrections, on a particular infinite family with leaves. It uses standard symmetry and Perron reductions. Wong's thesis already develops orbit reduction for stoquastic Hamiltonians, including ordinary double stars whose hubs are joined directly [7, Section 3.3, Theorem 13 and Example 2]. Our contribution is the full first-excitation ordering and uniform joint-limit expansion on the subdivided family below. It is not a new symmetry-reduction method. The comparisons concern the cited statements; no exhaustive priority claim or resolution of the universal conjecture is made.

2 The family and its full spectral gap

Let T_k have hubs a, b, a shared neighbor c, private leaves $u_1, \dots, u_k$ at a, and $v_1, \dots, v_k$ at b. Its only edges are

$$ac,\ bc,\ au_i,\ bv_i \quad (1 \le i \le k). \tag{4}$$

Then $n = 2k + 3$, $m = 2k + 2$, and the hubs have degree $k + 1$. Unlike an ordinary double star, their distance is two.

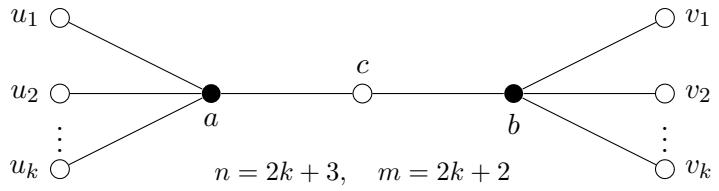


Figure 1: The tree T_k. Each hub has k private leaves, and the degree-two vertex c is their only shared neighbor. All displayed edges carry the same interaction in (1); the ellipses denote the remaining private leaves.

Put $q = s^2$, $\eta = 1 - q$, and $U = X_a X_b$. The operator

$$B = -U H_s(T_k) U \tag{5}$$

is positive semidefinite and preserves ordinary occupation number N in this transformed basis. Denote the largest eigenvalue of its *complete* number sector by λ_N. This definition does not restrict to permutation-symmetric states.

Theorem 1. *For every integer $k \ge 100$ and every real $\theta \in [1/2, 2]$, set $\eta = \theta/k$ and $s = \sqrt{1 - \eta}$. The full ground and first excited energies of (1) on T_k are simple and equal to $-\lambda_2$ and $-\lambda_3$, respectively. They satisfy*

$$\left| \frac{\Delta(H_s(T_k))}{\eta} - \left(1 + \frac{1}{2k} + \frac{1 - \theta}{4k^2}\right) \right| \le \frac{364}{k^3}. \tag{6}$$

In this whole domain $\Delta > \eta$, and Δ/η tends uniformly to one as $k \to \infty$. Every full sector $N \ge 4$ obeys

$$\lambda_N < \lambda_3 - \frac{(N - 3)\eta}{2}. \tag{7}$$

The original-gauge ground state is even and the first excitation is odd.

The proof has two distinct ingredients. Exact low-sector calculations identify λ_2, λ_3 with uniform errors. Full-space operator bounds then show that no other sector or internal ground-sector mode can compete with them. A difference of two chosen-sector trial energies would not suffice for the theorem.

2.1 Number convention and sector maxima

Each edge is oriented with its hub first. The hub flip maps its positive ket to $|10\rangle + s|01\rangle$, so the nonzero local matrix of B is

$$\begin{pmatrix} 1 & \sqrt{q} \\ \sqrt{q} & q \end{pmatrix}_{10,01}. \tag{8}$$

This is a unitary change of physical basis. Number conservation in this basis does not imply ordinary number conservation for H_s in the original basis. Two hub flips preserve the original parity $\bigotimes_i Z_i$.

For $q > 0$ and $0 < N < n$, the full N-occupation matrix is nonnegative and irreducible. Exchanges along the edges of a connected graph connect all N-subsets: holes can be moved along paths on a spanning tree to relocate tokens successively. Perron–Frobenius therefore gives a simple largest eigenvalue and a strictly positive vector in each such sector. Uniqueness makes that vector invariant under the two private-leaf permutation groups and under branch reflection. Hence the sector *maximum* can be computed in the leaf-symmetric, reflection-even block. No statement about an internal second eigenvector follows from this observation; those modes are bounded separately below.

3 Low-sector roots with uniform error control

3.1 Exact rational quotients

An occupation orbit is labeled by $(\alpha, \beta, \zeta, p, r)$, where $\alpha, \beta, \zeta \in \{0, 1\}$ are the hub and middle occupations, p, r are the private-leaf counts, and $\alpha + \beta + \zeta + p + r = N$. Write amplitudes on each orbit as $s^{\alpha+\beta}$ times a constant. This positive similarity gives a rational quotient Q_N in k, q. Its diagonal is

$$\begin{aligned} (Q_N)_{vv} &= f(\alpha, \zeta, p) + f(\beta, \zeta, r), \\ f(\alpha, \zeta, p) &= \alpha(1 - \zeta + k - p) + (1 - \alpha)q(\zeta + p). \end{aligned} \tag{9}$$

Its private-leaf row transitions are

$$(1, p) \longrightarrow (0, p + 1) : \; k - p, \qquad (0, p) \longrightarrow (1, p - 1) : \; qp, \tag{10}$$

with the other branch unchanged. For a hub–middle exchange the row coefficients are 1 from $(1, 0)$ and q from $(0, 1)$. Terms whose target occupancy is not allowed are absent. The positive diagonal metric is

$$w_v = \binom{k}{p}\binom{k}{r} q^{\alpha+\beta}, \qquad W Q_N = Q_N^{\mathsf{T}} W. \tag{11}$$

Reflection sends $(\alpha, \beta, \zeta, p, r)$ to $(\beta, \alpha, \zeta, r, p)$. For its even block, each row is summed over target reflection orbits; the orbit's cardinality multiplies its metric weight. The unfurled dimensions are 12 and 20 for $N = 2, 3$, and reflection reduces them to 7 and 11. Equations (9)–(11) specify every matrix entry without a fitted or numerical input. The positive metric also applies to auxiliary real $k \geq 100$, since the binomial products used here have $p, r \leq 3$.

3.2 Rational polynomial enclosures

Let $\chi_N(x;k,q)=\det(x\mathbb{I}-Q_N)$ for the reflection-even quotient. Set $e=1/k$, $q=1-\theta e$, and substitute

$$x=\frac{2}{e}+4+e(h-N\theta). \tag{12}$$

Remove the lowest Laurent power of e from the substituted determinant, calling the resulting polynomial $P_N(e,\theta,h)$. Direct exact calculation gives

$$P_2(0,\theta,h)=8h, \qquad P_3(0,\theta,h)=32h. \tag{13}$$

The polynomials, matrices and their exact rational coefficient bounds are supplied with the reproduction code.

The bounds use the box $0\le e\le 1/100$, $1/2\le\theta\le 2$, $|h|\le 1$. For a polynomial $F=\sum c_{ijl}e^i\theta^j h^l$, its coefficient norm

$$\mathcal{C}(F)=\sum_{i,j,l}|c_{ijl}|100^{-i}2^j \tag{14}$$

is a uniform upper bound for $|F|$ on this box. All coefficients and all operations in this norm are rational. Define the truncations

$$\bar{h}_2=-2e+(2\theta+10)e^2, \tag{15}$$

$$\bar{h}_3=-\frac{\theta+4}{2}e+\frac{\theta^2+7\theta+40}{4}e^2. \tag{16}$$

The exact determinant expansion cancels the residual terms of degrees zero, one and two. Applying (14) to the derivative deviation and to $P_N(e,\theta,\bar{h}_N)/e^3$ yields the outward rational bounds in Table 1.

N	$\deg_x\chi_N$	$\lvert\partial_h P_N-\ell_N\rvert$	$\lvert P_N(e,\theta,\bar{h}_N)\rvert/e^3$	C_N
2	7	$<187/200$	<572	81
3	11	$<1541/250$	<2597	101

Table 1: Uniform exact coefficient bounds, with $\ell_2=8$ and $\ell_3=32$. The last column bounds the root error by $|h-\bar{h}_N|\le C_N e^3$, equivalently an eigenvalue error $C_N e^4$. The tabulated inequalities round rational bounds outward; they are not estimates from floating-point sampling.

The derivative lower bounds are $1413/200$ and $6459/250$, respectively. The residual divided by the derivative lower bound is smaller than $81e^3$ and $101e^3$. Moreover $|\bar{h}_2|\le 107/5000$ and $|\bar{h}_3|\le 629/20000$ by the same norm. Adding $C_N/100^3$ keeps both error neighborhoods inside $|h|<1$. Strict monotonicity and the intermediate-value theorem thus give a unique simple real root in each enclosure.

It remains to identify these roots as the sector maxima. At $(k,\theta)=(100,1/2)$, exact rational root isolation counts all seven and eleven roots, including multiplicities, and identifies the enclosed roots as the largest simple roots. The second corner $(100,2)$ supplies an additional exact check. Their gap enclosures are reported in Table 2. These are complete quotient-root counts, rather than an iterative eigensolver's selected roots.

For auxiliary real $k\ge 100$, the positive metric (11) makes all eigenvalues real and continuous. The parameter domain is connected, and the enclosed root remains simple everywhere by the derivative bound. It cannot exchange order with another root without becoming multiple. The one exact anchor therefore identifies the largest roots on the whole domain. This real-k

continuation is algebraic; physical graphs still have integer k. Combining the enclosures with (12) gives

$$\left|\lambda_2 - \left(2k + 4 - \frac{2\theta}{k} - \frac{2}{k^2} + \frac{2\theta + 10}{k^3}\right)\right| \le \frac{81}{k^4}, \tag{17}$$

$$\left|\lambda_3 - \left(2k + 4 - \frac{3\theta}{k} - \frac{\theta + 4}{2k^2} + \frac{\theta^2 + 7\theta + 40}{4k^3}\right)\right| \le \frac{101}{k^4}. \tag{18}$$

θ	η	Complete root counts ($N = 2, 3$)	Δ/η enclosure
1/2	1/200	$(7, 11)$	$[1.005013181996, 1.005013181997]$
2	1/50	$(7, 11)$	$[1.004974283580, 1.004974283581]$

Table 2: Exact anchors at $k = 100$. Each characteristic polynomial has all roots isolated using rational intervals of width at most 10^{-24}, and its enclosed largest root is simple. The displayed gap-ratio intervals are rounded outward to twelve decimal places from the rational enclosures. Full-space ordering uses the analytic bounds in Section 4, rather than these finite checks.

4 Complete-sector ordering

4.1 A full-star operator inequality

Let M_L be the positive operator on a star with L leaves, using the oriented interaction (8), and let $N_\star$ be its occupation number. On the entire star Hilbert space,

$$M_L \le (L + 1)\mathbb{I} - \eta N_\star, \qquad \eta = 1 - q, \quad 0 \le q \le 1. \tag{19}$$

This is a consequence of the star spectrum, whose symmetric form is established in the source [1]. To apply it as an operator inequality we include every nonsymmetric copy and endpoint.

Decompose the L leaf qubits into spin $j = L/2 - d$, $0 \le d \le \lfloor L/2 \rfloor$, with multiplicity $\binom{L}{d} - \binom{L}{d-1}$, where $\binom{L}{-1} = 0$. Each copy has leaf occupations $p = d, \ldots, L - d$. In a paired total-N block, the hub/leaf states $(1, N - 1)$ and $(0, N)$ give

$$M_{d,N} = \begin{pmatrix} L - N + 1 & \sqrt{q(N - d)(L - d - N + 1)} \\ \sqrt{q(N - d)(L - d - N + 1)} & qN \end{pmatrix}. \tag{20}$$

This positive matrix has trace $L + 1 - \eta N$ and determinant $qd(L + 1 - d) \ge 0$. Each eigenvalue is consequently bounded above by the trace. The low singleton has hub occupation zero, $N = d$ and diagonal qN; its slack below the same bound is $L + 1 - N \ge 0$. The high singleton has hub occupation one, $N = L - d + 1$ and diagonal $L - N + 1$; its slack is $qN \ge 0$. These include the vacuum and fully occupied endpoints. The spin decomposition exhausts the full space with its multiplicities, proving (19). No irreducibility at $q = 0$ or $q = 1$ is needed.

4.2 Every higher number sector

The two $(k + 1)$-leaf stars in T_k overlap only at c. Their number operators add to $N + n_c$. Adding the two valid operator inequalities (19) gives

$$B \le (2k + 4)\mathbb{I} - \eta(N + n_c) \le (2k + 4)\mathbb{I} - \eta N. \tag{21}$$

This step does not require the overlapping stars to commute. It holds for every leaf-spin representation, reflection space and high-filling endpoint. Thus $\lambda_N \le 2k + 4 - \eta N$ for every complete sector.

Dropping the positive k^{-3} term from (18) gives

$$\lambda_3 \geq 2k + 4 - \frac{3\theta}{k} - \frac{3}{k^2} - \frac{101}{k^4}. \tag{22}$$

For $N \geq 4$, the difference between $\lambda_3 - (N-3)\eta/2$ and the sector envelope is at least

$$\frac{(N-3)\theta}{2k} - \frac{3}{k^2} - \frac{101}{k^4} \geq \frac{1}{k}\left(\frac{1}{4} - \frac{3}{100} - \frac{101}{100^3}\right) = \frac{219899}{1000000k} > 0. \tag{23}$$

This proves the stronger tail statement (7) on the full Hilbert space, rather than on its permutation-symmetric subspace.

4.3 Low sectors and internal ground-sector competitors

Write $B = A + C$, where A is the sum of the two disjoint private k-leaf stars and leaves c free, while C contains just ac, bc. The actual orientation of C has the hubs at a, b and the common site at c. Its full spectra in number sectors one and two are

$$\mathrm{spec}(C|_{N=1}) = \{0, 1, 1+2q\}, \qquad \mathrm{spec}(C|_{N=2}) = \{0, q, 2+q\}. \tag{24}$$

Its zero- and three-occupation blocks are zero, so $\|C\| = 2 + q$ for $0 \leq q \leq 1$.

The complete one-occupation spectrum of a private star is $k + q$ (simple), q (multiplicity $k - 1$) and zero. Its two-occupation maximum is $k - 1 + 2q$, simple for $q > 0$. In total $N = 2$ with $c = 0$, A has a unique maximum $2(k + q)$ from private fillings $(1, 1)$. Every other eigenvalue is at most $k + 2q$: this includes a nonsymmetric q leaf mode paired with $k + q$, and the $(2, 0)$ or $(0, 2)$ maxima $k - 1 + 2q$. For $c = 1$ the private filling is one, whose maximum is $k + q$. Thus the second largest eigenvalue of the *full* $N = 2$ block of A is at most $k + 2q$. Weyl's inequality then bounds every internal competitor of B in this sector by

$$\lambda_{\mathrm{second}}(B|_{N=2}) \leq k + 2q + \|C\| = k + 2 + 3q \leq k + 5. \tag{25}$$

Likewise $\lambda_1 \leq k + 2 + 2q \leq k + 4$, while $\lambda_0 = 0$. Equation (22) gives the uniform separation

$$\lambda_3 - (k + 5) \geq k - 1 - \frac{6}{k} - \frac{3}{k^2} - \frac{101}{k^4} \geq \frac{9893969899}{100000000} > 0. \tag{26}$$

All multiplicities and nonsymmetric private-leaf modes have therefore been retained and excluded from competition with λ_3.

4.4 Gap and simplicity

Subtract (18) from (17) and divide by $\eta = \theta/k$. The total error is at most $(81 + 101)/(\theta k^3) \leq 364/k^3$, giving (6). Its excess above one is bounded below by

$$\frac{\lambda_2 - \lambda_3}{\eta} - 1 \geq \frac{1}{2k} - \frac{1}{4k^2} - \frac{364}{k^3} \geq \frac{4611}{10000k} > 0. \tag{27}$$

In particular $\lambda_2 > \lambda_3$. The sector Perron argument, (23), (25) and (26) show that these are exactly the largest two eigenvalues of B in the complete physical space. Both are simple. Other levels inside $N = 3$ are strictly below its simple Perron maximum; those inside $N = 2$ obey (25). The sign change in (5) therefore gives $E_0 = -\lambda_2$ and $E_1 = -\lambda_3$ with multiplicity counted. Parity preservation by U makes the original ground even and the first excitation odd. The uniform error bound also proves the stated uniform limit. This completes Theorem 1.

5 Reproduction and physical interpretation

The exact supplementary calculation constructs the quotients from the source Pauli identity and collective-spin ladder factors, computes the two characteristic polynomials, evaluates rational coefficient norms, and isolates all anchor roots. Direct physical number-space matrices at $k = 3$, $s = 3/5$ intertwine exactly with the quotients on dimensions 36 and 84. These small systems check the transcription, rather than the universal parameter quantifiers. As a normalization and sign control, a direct local-ket construction agrees exactly with (2); the complete four-qubit star at $s = 1/2$ has $E_0 = -13/4$, $E_1 = -5/2$ and $\Delta = 3/4$, reproducing the known star law [1].

The calculation uses SymPy 1.14.0 for exact polynomial arithmetic and rational real-root isolation. No approximate eigensolver, fit or finite graph scan establishes the universal inequalities. AI assistance was used to develop and check the derivation, generate and review code, compare primary literature, and prepare the text. The mathematical basis of the result is the explicit full-space argument and reproducible exact algebra; the supplementary data contain the complete matrices, polynomials and rational enclosures.

The lowest excitation of this two-hub network is a number-three state above the number-two ground state in the hub-flipped basis. This is a physical parity statement in the original model, not ordinary original-gauge particle addition. Its energy scale is $\Delta \sim \theta/k$, with a positive relative excess $1/(2k)$ over the star threshold and a controlled θ-dependent next correction. Nonstar trees consequently approach the star ratio within this joint size/endpoint regime. The full-star envelope and the internal-sector estimate are what make this a prediction for the actual lowest excitation, including omitted symmetry modes.

The hub degree grows while each edge retains strength $J = 1$. The closing gap along $1 - s^2 = \theta/k$ is therefore not a bounded-degree thermodynamic limit, a phase transition, or an experimental realization claim. We establish neither an all-s result on T_k nor a theorem for general nonstar graphs, arbitrary weights or phases. General odd-first ordering is not settled. The original all-graph sharp-gap conjecture remains unresolved by this family calculation. Existing positive-gap and algorithmic results retain their separate scope.

Data and code availability

The local reproduction package contains the exact certificate program and its rational outputs, the complete manuscript sources, and regeneration instructions with the tested environment. The anchor table is generated from the saved rational enclosures. Primary papers are identified through bibliographic links; their full texts are not redistributed with the supplement.

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AI disclosure. OpenAI Codex was used for research ideation, derivation assistance, code generation, exact computational checks, literature retrieval and manuscript drafting. Its role extended beyond language editing. This local draft awaits the named author's critical scientific review, validation of the AI-assisted outputs, and confirmation of authorship and affiliation. Completed human validation is not asserted.

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